**Table C4. Fully unconstrained random parameters binary logit models with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), estimated separately for each year: multi-vehicle crashes without a motorcycle, Thailand, 2016–2023. Estimated parameter (*t*-statistic).**

| Variable description | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|
| **Constant [serious injury / fatal]** | | | | | | | | |
| Constant | –0.385 (–1.87) | 0.128 (0.72) | –0.712 (–3.59) | –0.984 (–5.91) | –1.079 (–6.74) | –0.940 (–4.83) | –1.266 (–4.88) | –0.526 (–2.61) |
| *Vehicle involvement* | | | | | | | | |
| One passenger car involved (1 if yes, 0 otherwise) | –0.541 (–4.54) | –0.934 (–6.96) | –0.562 (–4.48) | –0.404 (–3.37) | –0.510 (–4.40) | –0.825 (–5.96) | –0.585 (–4.55) | –0.671 (–5.37) |
| Two or more passenger cars involved (1 if yes, 0 otherwise) | –1.454 (–8.11) | –1.902 (–9.81) | –1.162 (–6.26) | –0.856 (–5.14) | –1.217 (–6.68) | –1.769 (–7.85) | –1.206 (–6.08) | –1.534 (–7.25) |
| One pickup truck involved (1 if yes, 0 otherwise) | –0.282 (–2.30) | –0.556 (–4.29) | –0.410 (–3.37) | –0.752 (–3.55) | –0.543 (–4.39) | –0.726 (–5.11) | –0.354 (–2.61) | –0.297 (–2.31) |
| *(Standard deviation of parameter distribution)* | | | | 1.048 (2.84) | | | | |
| Two or more pickup trucks involved (1 if yes, 0 otherwise) | –0.751 (–4.56) | –1.897 (–3.61) | –1.934 (–3.94) | –1.046 (–6.53) | –1.128 (–6.96) | –1.456 (–7.95) | –1.068 (–6.17) | –1.578 (–3.87) |
| *(Standard deviation of parameter distribution)* | | 2.240 (2.94) | 2.112 (3.57) | | | | | 1.646 (3.50) |
| One 10-wheel truck involved (1 if yes, 0 otherwise) | | | –1.886 (–1.76) | | | | | |
| *(Standard deviation of parameter distribution)* | | | 3.703 (2.43) | | | | | |
| One truck with more than 10 wheels involved (1 if yes, 0 otherwise) | 0.258 (1.89) | | 0.453 (3.40) | | | | –0.646 (–1.68) | 0.367 (2.77) |
| *(Standard deviation of parameter distribution)* | | | | | | | 1.925 (3.45) | |
| Two or more trucks with more than 10 wheels involved (1 if yes, 0 otherwise) | | –0.946 (–3.35) | | –0.565 (–2.07) | –0.609 (–1.92) | | | |
| Van involved (1 if yes, 0 otherwise) | | | –0.340 (–1.76) | | | | | |
| *Crash type* | | | | | | | | |
| Rear-end collision in the same lane (1 if yes, 0 otherwise) | –0.427 (–3.71) | | –0.231 (–1.84) | –0.439 (–3.57) | –0.448 (–3.84) | | –0.341 (–2.24) | –0.404 (–2.74) |
| Sideswipe (parallel movement) (1 if yes, 0 otherwise) | –0.655 (–3.48) | –0.565 (–2.60) | –0.653 (–3.22) | –0.974 (–4.48) | | | –2.806 (–2.42) | –0.505 (–2.67) |
| *(Standard deviation of parameter distribution)* | | | | | | | 3.431 (3.12) | |
| Head-on collision (1 if yes, 0 otherwise) | 0.770 (3.14) | 1.379 (5.11) | 1.115 (6.35) | 1.418 (7.06) | 1.312 (8.56) | 1.246 (7.35) | 0.791 (4.49) | 0.891 (4.90) |
| *Crash cause* | | | | | | | | |
| Speeding (1 if yes, 0 otherwise) | –0.309 (–2.27) | | | | | | 0.542 (2.72) | |
| Cutting in too closely (1 if yes, 0 otherwise) | –1.838 (–2.19) | –0.336 (–1.83) | | –0.278 (–1.66) | | | 0.586 (2.63) | –0.489 (–2.71) |
| *(Standard deviation of parameter distribution)* | 2.882 (2.68) | | | | | | | |
| Driver fell asleep (1 if yes, 0 otherwise) | | 0.893 (3.72) | 0.638 (2.95) | | 0.874 (3.70) | | 1.104 (3.78) | 0.477 (2.03) |
| *Roadway characteristics* | | | | | | | | |
| Crash on the main carriageway (1 if yes, 0 otherwise) | –0.536 (–2.77) | –0.484 (–2.54) | | | | 0.488 (2.07) | | |
| Road under construction (1 if yes, 0 otherwise) | 0.481 (1.87) | | | 1.180 (3.55) | | | 0.558 (2.24) | |
| Painted median (1 if yes, 0 otherwise) | 0.419 (1.81) | | | 0.534 (2.96) | 0.319 (1.86) | | | |
| Raised or earth median (1 if yes, 0 otherwise) | –0.577 (–3.84) | –0.523 (–3.05) | –0.582 (–3.77) | | | –0.384 (–2.50) | –0.282 (–1.95) | –0.688 (–4.72) |
| Depressed median (1 if yes, 0 otherwise) | –0.343 (–2.51) | –0.517 (–3.53) | –0.720 (–4.77) | | | –0.919 (–2.16) | –0.480 (–2.95) | –1.639 (–2.93) |
| *(Standard deviation of parameter distribution)* | | | | | | 2.408 (3.83) | | 2.659 (3.55) |
| Barrier median (1 if yes, 0 otherwise) | –1.246 (–6.38) | –1.364 (–7.15) | –0.874 (–4.78) | –1.276 (–7.81) | –1.347 (–7.89) | –1.498 (–7.98) | –2.106 (–10.00) | –1.850 (–10.82) |
| Horizontal curve (1 if yes, 0 otherwise) | 0.497 (3.09) | | | –1.283 (–2.10) | | | | –0.356 (–1.92) |
| *(Standard deviation of parameter distribution)* | | | | 3.718 (3.49) | | | | |
| Concrete pavement surface (1 if yes, 0 otherwise) | | –0.418 (–2.19) | –0.316 (–1.83) | | | | | |
| On grade (1 if yes, 0 otherwise) | –0.773 (–2.42) | | | | | | | |
| Four-leg intersection (1 if yes, 0 otherwise) | –0.514 (–2.42) | | 0.601 (2.80) | | | | | |

| Variable description | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|
| U-turn with deceleration lane only (1 if yes, 0 otherwise) | | | 1.057 (3.01) | | | | | 0.582 (1.82) |
| U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise) | 0.587 (1.76) | | | | | | | |
| *Environmental characteristics* | | | | | | | | |
| Wet road surface (1 if yes, 0 otherwise) | –0.332 (–1.90) | | –0.658 (–3.75) | | | –0.410 (–2.10) | –0.422 (–2.42) | –0.760 (–3.81) |
| Raining (1 if yes, 0 otherwise) | | | | | –0.482 (–2.90) | | | |
| Dark, no street lighting (1 if yes, 0 otherwise) | 0.463 (2.67) | 0.992 (4.52) | 1.330 (7.96) | 1.372 (7.33) | 0.993 (4.77) | | 0.493 (2.48) | 0.684 (3.98) |
| Dark, with street lighting (1 if yes, 0 otherwise) | | 0.729 (4.64) | 0.515 (4.35) | 0.449 (3.76) | 0.829 (5.89) | | | |
| *Temporal characteristics* | | | | | | | | |
| Songkran festival period (1 if yes, 0 otherwise) | | | | | | 0.561 (2.16) | –0.437 (–1.75) | |
| New Year festival period (1 if yes, 0 otherwise) | –0.423 (–2.39) | | | –0.735 (–3.52) | | | | |
| Evening (1 if yes, 0 otherwise) | 0.600 (4.57) | –0.458 (–2.74) | | | –0.932 (–3.26) | 0.642 (4.63) | 0.434 (3.12) | |
| *(Standard deviation of parameter distribution)* | | | | | 1.132 (2.56) | | | |
| Midnight / early morning (1 if yes, 0 otherwise) | 0.769 (5.50) | | | 0.384 (2.84) | | 0.874 (5.28) | 0.444 (2.85) | 0.533 (3.97) |
| Peak hours (1 if yes, 0 otherwise) | | –0.274 (–1.67) | | | –0.644 (–1.72) | | | |
| *(Standard deviation of parameter distribution)* | | | | | 1.906 (4.02) | | | |
| *Region* | | | | | | | | |
| Eastern region (1 if yes, 0 otherwise) | 0.618 (3.80) | | 0.680 (3.78) | | 0.442 (2.66) | 0.534 (3.18) | 0.752 (4.62) | 0.562 (3.93) |
| Northeastern region (1 if yes, 0 otherwise) | 0.694 (5.05) | –0.306 (–1.10) | 0.485 (3.54) | 0.740 (5.36) | 0.746 (5.63) | 0.093 (0.24) | 0.753 (4.87) | 0.522 (3.62) |
| *(Standard deviation of parameter distribution)* | | 1.816 (3.08) | | | | 1.902 (2.72) | | |
| Northern region (1 if yes, 0 otherwise) | | | | 0.404 (1.90) | | –0.391 (–1.73) | –0.427 (–2.01) | |
| Southern region (1 if yes, 0 otherwise) | 0.304 (1.85) | | 0.531 (3.14) | 0.436 (2.59) | 0.637 (4.06) | 0.331 (1.77) | 0.327 (1.80) | |
| Western region (1 if yes, 0 otherwise) | 0.581 (3.08) | | 0.403 (2.03) | 0.600 (2.99) | 1.042 (5.77) | 0.874 (4.25) | 0.428 (1.94) | 0.737 (3.47) |
| **Heterogeneity in the means of random parameters** | | | | | | | | |
| Two or more pickup trucks involved: One 6-wheel truck involved | | | | | | | | 1.845 (2.93) |
| Two or more pickup trucks involved: Two or more trucks with more than 10 wheels involved | | | 3.621 (1.89) | | | | | |
| Two or more pickup trucks involved: Rear-end collision in the same lane | | –1.224 (–2.37) | | | | | | |
| One 10-wheel truck involved: Horizontal curve | | | 2.672 (1.91) | | | | | |
| One truck with more than 10 wheels involved: Van involved | | | | | | | 2.017 (2.64) | |
| One truck with more than 10 wheels involved: Concrete pavement surface | | | | | | | –0.978 (–2.00) | |
| Cutting in too closely: T-intersection | 1.947 (2.12) | | | | | | | |
| Depressed median: Van involved | | | | | | | | 1.392 (2.23) |
| Depressed median: Raining | | | | | | –1.505 (–2.29) | | |
| Evening: One truck with more than 10 wheels involved | | | | | 0.999 (3.44) | | | |
| Evening: New Year festival period | | | | | –1.314 (–1.90) | | | |
| Northeastern region: Dark, no street lighting | | | | | | 2.027 (2.81) | | |
| **Heterogeneity in the variances of random parameters** | | | | | | | | |
| One truck with more than 10 wheels involved: Dark, with street lighting | | | | | | | 0.502 (2.28) | |

| Variable description | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|
| Evening: Depressed median | | | | | 0.660 (1.96) | | | |
| Northeastern region: Four-leg intersection | | 1.019 (1.81) | | | | | | |
| Northeastern region: Weekend | | | | | | 0.778 (2.16) | | |
| **Model statistics** | | | | | | | | |
| Number of observations | 3,509 | 3,217 | 3,505 | 4,281 | 4,933 | 4,189 | 4,647 | 5,365 |
| Number of estimated parameters | 32 | 24 | 29 | 24 | 25 | 24 | 31 | 26 |
| Random parameters / heterogeneity in means / in variances | 1 / 1 / 0 | 2 / 1 / 1 | 2 / 2 / 0 | 2 / 0 / 0 | 2 / 2 / 1 | 2 / 2 / 1 | 2 / 2 / 1 | 2 / 2 / 0 |
| Log-likelihood at zero, LL(0) | –2432.25 | –2229.85 | –2429.48 | –2967.36 | –3419.30 | –2903.59 | –3221.05 | –3718.73 |
| Log-likelihood with constant only, LL(C) | –1746.44 | –1617.53 | –1786.67 | –2003.44 | –2143.38 | –1797.65 | –1891.68 | –2106.07 |
| Log-likelihood at convergence, LL(β) | –1426.65 | –1365.03 | –1527.42 | –1648.82 | –1767.71 | –1505.41 | –1563.36 | –1737.34 |
| McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$ | 0.413 | 0.388 | 0.371 | 0.444 | 0.483 | 0.482 | 0.515 | 0.533 |

**Notes.** A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Each column is a separate model for one year. Each model allows at most two random parameters, two heterogeneity-in-means terms and one heterogeneity-in-variance term, searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level. A blank cell means the variable is not in that year's model. Rows with a standard deviation give the mean of the random parameter in the years where it is random and a fixed parameter in the other years. A variable used in a heterogeneity function enters only the random parameter shown in that year and is removed from the main utility. Random parameters normally distributed; 2,000 Halton draws.